Z-TRANSFORM

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Introduction

The sequences $\{u_n\}$ that are to be Z-transformed, must be defined for $n \geq 0$. However, the values of u_n for $n < 0$, if any, do not effect the Z-transform. It is noteworthy that Z-transforms have properties quite similar to Laplace transforms. For almost every result in Laplace transform there is a corresponding result in z-transform.

Definition(Z-transform)

The Z-transform of a sequence u_n defined for discrete values $n = 0, 1, 2, \ldots$ (and $u_n = 0$ for $n < 0$) is denoted by $Z(u_n)$ or $U(z)$ and is defined as

$$Z(u_n) = \sum_{n=0}^{\infty} u_n z^{-n} = U(z) \tag{1}$$

where U is a function of z. The Z-transform of a sequence u_n exists only when the infinite series in (1) is convergent.

The Inverse Z-transform is denoted by $Z^{-1}(U(z)) = u_n$ determines the sequence u_n which generates the given Z-transform.

Existence of Z-transform

Result

Let $\{u_n\}$ be a sequence of exponential form, that is, there exist numbers $N > 0$, $t_0 > 0$ and $n_0 > 0$ such that

$$|u_n| < Ne^{nt_0},$$

for all $n \geq n_0$, then the Z-transform of u_n exists. Therefore, every sequence $\{u_n\}$ of exponential order has Z-transform. This is a necessary as well as sufficient condition for the existence of Z-transform.

Z-transform of some standard sequences (or functions)

Example

Show that $Z(1) = \frac{z}{z-1}$.

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Solution

By definition

$$Z(1) = \sum_{n=0}^{\infty} z^{-n} = 1 + \frac{1}{z} + \frac{1}{z^2} + \cdots = \frac{1}{1 - 1/z} = \frac{z}{z-1}$$

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Recurrence formula

Example(Recurrence formula)

Show that $Z(n^p) = -z\frac{d}{dz}Z(n^{p-1})$, p being a positive integer.

Solution

By definition

$$Z(n^p) = \sum_{n=0}^{\infty} n^p z^{-n} \tag{2}$$

Replacing p by $p-1$, we get

$$Z(n^{p-1}) = \sum_{n=0}^{\infty} n^{p-1} z^{-n} \tag{3}$$

Differentiating (3) with respect to z, we get

$$\begin{aligned}
\frac{d}{dz}\left[Z(n^{p-1})\right] &= \frac{d}{dz}\left[\sum_{n=0}^{\infty} n^{p-1} z^{-n}\right] = \sum_{n=0}^{\infty} n^{p-1}(-n) z^{-n-1} \\
&= -z^{-1} \sum_{n=0}^{\infty} n^p z^{-n} \\
&= -z^{-1} Z(n^p) \qquad \text{using (2)}
\end{aligned}$$

Thus

$$Z(n^p) = -z \frac{d}{dz} Z(n^{p-1}).$$

In particular, we have the following formulae

- $Z(1) = \dfrac{z}{z-1}$
- $Z(n) = \dfrac{z}{(z-1)^2}$
- $Z(n^2) = \dfrac{z^2+z}{(z-1)^3}$
- $Z(n^3) = \dfrac{z^3+4z^2+z}{(z-1)^4}$
- $Z(n^4) = \dfrac{z^4+11z^3+11z^2+z}{(z-1)^5}$

Example

Show that $Z\left(\frac{1}{n\,!}\right) = e^{\frac{1}{z}}$.

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Solution

By definition

$$Z\left(\frac{1}{n\,!}\right) = \sum_{n=0}^{\infty} \frac{1}{n\,!} z^{-n} = 1 + \frac{1}{z} + \frac{1}{2!}\frac{1}{z^2} + \frac{1}{3!}\frac{1}{z^3} + \cdots = e^{\frac{1}{z}}$$

Example

Show that the Z-transform of unit step function is $\frac{z}{z-1}$, i.e., show that

$$Z(u_n) = \frac{z}{z-1},$$

where $u_n = \begin{cases} 1 & \text{if } n \geq 0 \\ 0 & \text{if } n < 0 \end{cases}$.

Example

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where $u_n = \begin{cases} 1 & \text{if } n \geq 0 \\ 0 & \text{if } n < 0 \end{cases}$.

Solution

By definition

$$Z(u_n) = \sum_{n=0}^{\infty} u_n z^{-n} = \sum_{n=0}^{\infty} 1 \cdot z^{-n} = 1 + \frac{1}{z} + \frac{1}{z^2} + \cdots = \frac{1}{1 - 1/z} = \frac{z}{z-1}$$

The Z-transform is analogous to Laplace transform

Since the Laplace transform of a continuous(or piecewise continuous) function $f(t)$ is defined as $\int_0^\infty f(t)e^{-st}dt$, where $s > 0$ and $t \geq 0$, on similar lines, Z-transform of the function u_n is defined for discrete values $n = 0, 1, 2\ldots$ better known as sequence (u_n) where $u_n = 0$ for $n < 0$, as $\sum_{n=0}^{\infty} u_n z^{-n}$ provided it converges. There are following fundamental differences in Laplace transform and Z-transform

Sr. No.	**Laplace transform**	**Z-transform**
1	The Laplace transform definition involves an integral	The z-transform definition involves a summation
2	Applying the Laplace transform to certain ordinary differential equations turns them into simpler (algebraic) equations	The z-transform converts certain difference equations to algebraic equations
3	Use of the Laplace transform gives rise to the basic concept of the transfer function of a continuous (or analog) system.	Use of the z-transform gives rise to the concept of the transfer function of discrete (or digital) systems.

Properties of Z-transform

Linearity property

If u_n, v_n are Z-transformable functions and a, b are any two constants then

$$Z(au_n + bv_n) = aZ(u_n) + bZ(v_n).$$

Properties of Z-transform

Linearity property

If u_n, v_n are Z-transformable functions and a, b are any two constants then

$$Z(au_n + bv_n) = aZ(u_n) + bZ(v_n).$$

Change of scale or damping rule

If $Z(u_n) = U(z)$ then $Z(a^{-n}u_n) = U(az)$. In particular, $Z(a^n u_n) = U\left(\frac{z}{a}\right)$.

Some standard results (using damping rule)

The application of damping rule leads to the following standard results:

Example

Show that

(i) $Z(na^n) = \dfrac{az}{(z-a)^2}$

(ii) $Z(n^2a^n) = \dfrac{az^2 + a^2z}{(z-a)^3}$

(iii) $Z(\cos n\theta) = \dfrac{z(z - \cos\theta)}{z^2 - 2z\cos\theta + 1}$ and $Z(\sin n\theta) = \dfrac{z\sin\theta}{z^2 - 2z\cos\theta + 1}$

(iv) $Z(a^n \cos n\theta) = \dfrac{z(z - a\cos\theta)}{z^2 - 2az\cos\theta + a^2}$ and $Z(a^n \sin n\theta) = \dfrac{az\sin\theta}{z^2 - 2az\cos\theta + a^2}$.

Solution

(i) We know that $Z(n) = \frac{z}{(z-1)^2}$. Applying damping rule, we get

$$Z(na^n) = U(a^{-1}z) = \frac{a^{-1}z}{(a^{-1}z-1)^2} = \frac{az}{(z-a)^2}.$$

(ii) We know that $Z(n^2) = \frac{z^2+z}{(z-1)^3}$. Applying damping rule, we have

$$Z(n^2a^n) = U(a^{-1}z) = \frac{(a^{-1}z)^2 + a^{-1}z}{(a^{-1}z-1)^3} = \frac{a(z^2+az)}{(z-a)^3}.$$

(iii) We know that $Z(1) = \frac{z}{z-1}$. Applying damping rule yields

$$Z(e^{-in\theta}) = Z((e^{i\theta})^{-n} \cdot 1) = \frac{ze^{i\theta}}{ze^{i\theta}-1} = \frac{z}{z-e^{-i\theta}} = \frac{z}{[(z-\cos\theta)+i\sin\theta]}$$

$$= \frac{z[(z-\cos\theta)-i\sin\theta]}{[(z-\cos\theta)+i\sin\theta][(z-\cos\theta)-i\sin\theta]}$$

$$= \frac{z^2-z\cos\theta}{z^2-2z\cos\theta+1} - \frac{\sin\theta}{z^2-2z\cos\theta+1}$$

Solution cont...

Thus

$$Z(\cos\theta - i\sin\theta) = \frac{z^2 - z\cos\theta}{z^2 - 2z\cos\theta + 1} - \frac{\sin\theta}{z^2 - 2z\cos\theta + 1}$$

Applying Linearity property yields

$$Z(\cos n\theta) - i(\sin n\theta) = \frac{z^2 - z\cos\theta}{z^2 - 2z\cos\theta + 1} - \frac{\sin\theta}{z^2 - 2z\cos\theta + 1}$$

Equating real and imaginary parts, we get

$$Z(\cos n\theta) = \frac{z(z - \cos\theta)}{z^2 - 2z\cos\theta + 1}.$$

and

$$Z(\sin n\theta) = \frac{z\sin\theta}{z^2 - 2z\cos\theta + 1}.$$

Solution cont. . .

(iv) We know that $Z(\cos n\theta) = \dfrac{z(z-\cos\theta)}{z^2-2z\cos\theta+1}$ and $Z(\sin n\theta) = \dfrac{z\sin\theta}{z^2-2z\cos\theta+1}$.
Applying damping rule, we get

$$Z(a^n\cos n\theta) = \frac{a^{-1}z(a^{-1}z-\cos\theta)}{(a^{-1}z)^2-2a^{-1}z\cos\theta+1} = \frac{z(z-a\cos\theta)}{z^2-2az\cos\theta+a^2}$$

$$\implies Z(a^n\cos n\theta) = \frac{z(z-a\cos\theta)}{z^2-2az\cos\theta+a^2}$$

and

$$Z(a^n\sin n\theta) = \frac{a^{-1}z\sin\theta}{(a^{-1}z)^2-2a^{-1}z\cos\theta+1} = \frac{az\sin\theta}{z^2-2az\cos\theta+a^2}$$

$$\implies Z(a^n\sin n\theta) = \frac{az\sin\theta}{z^2-2az\cos\theta+a^2}$$

Shifting property

(a) Shifting u_n to the right: If $Z(u_n) = U(z)$ then

$$Z(u_{n-k}) = z^{-k}U(z)$$

for $k > 0$.

(b) Shifting u_n to the left: If $Z(u_n) = U(z)$ then

$$Z(u_{n+k}) = z^k \left[U(z) - u_0 - \frac{u_1}{z} - \frac{u_2}{z^2} - \cdots - \frac{u_{k-1}}{z^{k-1}}\right].$$

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$$Z(u_{n+k}) = z^k \left[U(z) - u_0 - \frac{u_1}{z} - \frac{u_2}{z^2} - \cdots - \frac{u_{k-1}}{z^{k-1}}\right].$$

In Particular, we have

- $Z(u_{n+1}) = z[U(z) - u_0]$
- $Z(u_{n+2}) = z^2[U(z) - u_0 - u_1 z^{-1}]$
- $Z(u_{n+3}) = z^3[U(z) - u_0 - u_1 z^{-1} - u_2 z^{-2}]$.

Example

Show that $Z\left(\frac{1}{n!}\right) = e^{1/z}$. Hence evaluate $Z\left(\frac{1}{(n+1)!}\right)$ and $Z\left(\frac{1}{(n+2)!}\right)$.

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Solution

We have

$$Z\left(\frac{1}{n!}\right) = \sum_{n=0}^{\infty} \frac{1}{n!} z^{-n} = 1 + \frac{1}{1!}\frac{1}{z} + \frac{1}{2!}\frac{1}{z^2} + \frac{1}{3!}\frac{1}{z^3} + \cdots = e^{1/z}$$

Shifting $u_n = \frac{1}{n!}$ one unit to the left gives

$$Z\left(\frac{1}{(n+1)!}\right) = z\left[Z\left(\frac{1}{n!}\right) - 1\right] = z(e^{1/z} - 1).$$

Similarly, Shifting $u_n = \frac{1}{n!}$ two units to the left gives

$$Z\left(\frac{1}{(n+2)!}\right) = z^2(e^{1/z} - 1 - z^{-1}).$$

Multiplication by n property

- If Z-transform of u_n is $U(z)$, i.e., $Z(u_n) = U(z)$ then

$$Z(nu_n) = -z\frac{dU(z)}{dz}$$

- If Z-transform of u_n is $U(z)$, i.e., $Z(u_n) = U(z)$ then

$$Z(n^p u_n) = (-z)^p \frac{d^p U(z)}{dz^p}$$

Example

Find the Z-transform of

1. $n \sin n\theta$
2. $n^2 e^{n\theta}$

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Solution

We know that $Z(nu_n) = -z\frac{dU(z)}{dz}$ and $Z(\sin n\theta) = \frac{z \sin\theta}{z^2 - 2z\cos\theta + 1}$. Therefore

$$\begin{aligned} Z(n \sin n\theta) &= -z\frac{d}{dz}[Z(\sin n\theta)] \\ &= -z\frac{d}{dz}\left[\frac{z \sin\theta}{z^2 - 2z\cos\theta + 1}\right] \\ &= \frac{z(z^2-1)\sin\theta}{(z^2 - 2z\cos\theta + 1)^2} \end{aligned}$$

Solution cont. . .

We know that $Z(e^{n\theta}) = \frac{z}{z-e^{\theta}}$. Therefore

$$\begin{aligned} Z(n^2 e^{n\theta}) &= (-z)^2 \frac{d^2}{dz^2}\left[Z(e^{n\theta})\right] \\ &= z^2 \frac{d^2}{dz^2}\left[\frac{z}{z-e^{\theta}}\right] \\ &= \frac{z(z+e^{\theta})e^{\theta}}{(z-e^{\theta})^3} \end{aligned}$$

Two Basic Theorems

Initial and Final value theorem

In applications, we often need the values of u_n for $n = 0$ or as $n \to \infty$ without requiring complete knowledge of u_n. The Initial value and final value theorems determine the values of u_n for $n = 0$ and for $n \to \infty$ without the complete knowledge of u_n. The precise relationship is given by the following initial and final value theorems:

- **Initial value theorem:** If $Z(u_n) = U(z)$, then $u_0 = \lim_{z\to\infty} U(z)$.

- **Final value theorem:** If $Z(u_n) = U(z)$, then

$$\lim_{n\to\infty}(u_n) = \lim_{z\to 1}(z-1)U(z).$$

Proof of initial value theorem

We know that

$$U(z) = Z(u_n) = \sum_{n=0}^{\infty} u_n z^{-n} = u_0 + \frac{u_1}{z} + \frac{u_2}{z^2} + \cdots$$

Taking limits as $z \to \infty$, we have

$$\lim_{z\to\infty} U(z) = u_0.$$

Similarly, additional initial values can be found successively, giving

$$u_1 = \lim_{z\to\infty} z[U(z) - u_0]$$
$$u_2 = \lim_{z\to\infty} z^2 \left[U(z) - u_0 - \frac{u_1}{z}\right]$$
$$u_3 = \lim_{z\to\infty} z^3 \left[U(z) - u_0 - \frac{u_1}{z} - \frac{u_2}{z^2}\right]$$

and so on.

Example

If $U(z) = \dfrac{2z^2 + 5z + 14}{(z-1)^4}$, evaluate u_2 and u_3.

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Solution

By initial value theorem

$$u_0 = \lim_{z\to\infty} U(z) = \lim_{z\to\infty} \frac{2z^2 + 5z + 14}{(z-1)^4} = 0.$$

Similarly,

$$u_1 = \lim_{z\to\infty} z[U(z) - u_0] = \lim_{z\to\infty} \frac{2z^2 + 5z + 14}{(z-1)^4} = 0.$$

$$u_2 = \lim_{z\to\infty} z^2 \left[U(z) - u_0 - \frac{u_1}{z}\right] = \lim_{z\to\infty} z^2 \frac{2z^2 + 5z + 14}{(z-1)^4} = 2$$

$$u_3 = \lim_{z\to\infty} z^3 \left[U(z) - u_0 - \frac{u_1}{z} - \frac{u_2}{z^2}\right] = \lim_{z\to\infty} z^3 \left[U(z) - \frac{2}{z^2}\right] = 13$$

Table 1: STANDARD Z-TRANSFORMS

Sr. No.	Sequence: u_n (for $n \geq 0$)	$U(z)$: Z-transform
1	n	$\frac{z}{(z-1)^2}$
2	n^2	$\frac{z^2+z}{(z-1)^3}$
3	n^p (Recurrence formula)	$-z\frac{d}{dz}[Z(n^{p-1}]$
4	a^n	$\frac{z}{z-a}$
5	nu_n (Multiplication by n)	$-z\frac{d}{dz}[Z(u_n)]$
6	$a^n u_n$ (Damping rule)	$U\left(\frac{z}{a}\right)$
7	u_{n+1}	$z[U(z) - u_0]$
8	u_{n+2}	$z^2\left[U(z) - u_0 - \frac{u_1}{z}\right]$
9	u_{n+3}	$z^3\left[U(z) - u_0 - \frac{u_1}{z} - \frac{u_2}{z^2}\right]$
10	u_{n-k}	$z^{-k}Z(u_n)$
11	$\sin n\theta$	$\frac{z\sin\theta}{z^2-2z\cos\theta+1}$
12	$\sin n\theta$	$\frac{z(z-\cos\theta)}{z^2-2z\cos\theta+1}$

Example

Find the Z-transform of $2n + 5\sin\frac{n\pi}{4} - 3a^4$.

Example

Find the Z-transform of $2n + 5\sin\frac{n\pi}{4} - 3a^4$.

Solution

By linearity property

$$Z\left(2n + 5\sin\frac{n\pi}{4} - 3a^4\right) = 2Z(n) + 5Z\left(\sin\frac{n\pi}{4}\right) - 3a^4 Z(1)$$

$$= \frac{2z}{(z-1)^2} + \frac{5z\sin\frac{\pi}{4}}{z^2 - 2z\cos\frac{\pi}{4} + 1} - 3a^4\frac{z}{z-1}$$

$$= \frac{2z}{(z-1)^2} + \frac{5z\frac{1}{\sqrt{2}}}{z^2 - \sqrt{2}z + 1} - \frac{3a^4 z}{z-1}$$

Example

Find $Z[(\cos\theta + i\sin\theta)^n]$.

Example

Find $Z[(\cos\theta + i\sin\theta)^n]$.

Solution

$$\begin{aligned} Z[(\cos\theta + i\sin\theta)^n] &= Z[(e^{i\theta})^n] \\ &= Z(e^{in\theta}) \\ &= \frac{z}{z - e^{i\theta}} \end{aligned}$$

Since $Z(a^n) = \frac{z}{z-a}$.

Example

Find $Z[(n+1)^2]$.

Example

Find $Z[(n+1)^2]$.

Solution

$$\begin{aligned} Z[(n+1)^2] &= Z(n^2+2n+1) \\ &= Z(n^2)+2Z(n)+Z(1) \qquad \text{[Linearity property]} \\ &= \frac{z^2+z}{(z-1)^3}+\frac{2z}{(z-1)^2}+\frac{z}{z-1} \\ &= \frac{z^3+z^2}{(z-1)^3} \end{aligned}$$

Another Method(Using shifting property): We know that $Z(n^2)=\frac{z^2+z}{(z-1)^3}$ and $Z(u_{n+k})=z^k\left[U(z)-u_0-\frac{u_1}{z}-\frac{u_2}{z^2}-\cdots-\frac{u_{k-1}}{z^{k-1}}\right]$. Therefore

$$Z[(n+1)^2]=z[Z(n^2)-0]=\frac{z^3+z^2}{(z-1)^3}.$$

Example

Find $Z(e^{-an}\sin n\theta)$

Example

Find $Z(e^{-an}\sin n\theta)$

Solution

We know that $Z(\sin n\theta) = \dfrac{z\sin\theta}{Z^2 - 2z\cos\theta + 1}$. Now

$$Z(e^{-an}\sin n\theta) = Z[(e^a)^{-n}\sin n\theta].$$

Applying damping rule $Z[a^{-n}u_n] = U(az)$, we get

$$Z(e^{-an}\sin n\theta) = \frac{e^a z\sin\theta}{(e^a z)^2 - 2e^a z\cos\theta + 1} = \frac{e^a z\sin\theta}{e^{2a}z^2 - 2e^a z\cos\theta + 1}.$$

Example

Find $Z(n\cos n\theta)$.

Example

Find $Z(n\cos n\theta)$.

Solution

We know that

$$Z(nu_n) = -z\frac{dU(z)}{dz}.$$

Here $u_n = \cos n\theta$ and

$$Z(u_n) = U(z) = \frac{z(z-\cos\theta)}{z^2 - 2z\cos\theta + 1}$$

Therefore

$$\begin{aligned} Z(n\cos n\theta) &= -z\frac{d}{dz}\left(\frac{z(z-\cos\theta)}{z^2 - 2z\cos\theta + 1}\right) \\ &= \frac{z^3\cos\theta - 2z^2 + z\cos\theta}{(z^2 - 2z\cos\theta + 1)^2} \end{aligned}$$

Example

Find $Z(u_{n+2})$ if $Z(u_n) = \frac{z}{z-1} + \frac{z}{z^2+1}$.

Example

Find $Z(u_{n+2})$ if $Z(u_n) = \frac{z}{z-1} + \frac{z}{z^2+1}$.

Solution

By shifting theorem, We know that

$$Z(u_{n+2}) = z^2 \left[U(z) - u_0 - \frac{u_1}{z} \right].$$

Here, we need to compute u_0 and u_1. By initial value theorem, we have

$$u_0 = \lim_{z\to\infty} U(z) = \lim_{z\to\infty} \left[\frac{z}{z-1} + \frac{z}{z^2+1} \right] = 1$$

$$u_1 = \lim_{z\to\infty} z[U(z) - u_0] = \lim_{z\to\infty} z \cdot \frac{2z^2 - z + 1}{(z-1)(z^2+1)} = 2$$

Therefore

$$Z(u_{n+2}) = z^2 \left[\frac{z}{z-1} + \frac{z}{z^2+1} - 1 - \frac{2}{z} \right] = \frac{z(z^2 - z + 2)}{(z-1)(z^2+1)}$$

Exercise 1:

Find the Z-transform of the following functions

(i) $3n - 4\sin\left(\frac{n\pi}{4}\right) + 5a$ **Ans.** $\dfrac{(3-5a)z + 5az^2}{(z-1)^2} - \dfrac{2\sqrt{2}z}{z^2 - \sqrt{2}z + 1}$

(ii) $(n+1)^2$ **Ans.** $\dfrac{z^2(2z+1)}{(z-1)^3}$

(iii) $\sin(3n+5)$ **Ans.** $\dfrac{z^2\sin 5 - z\sin 2}{z^2 - 2z\cos 3 + 1}$

Exercise 2:

Find the Z-transforms of the following

(i) e^{an} **Ans.** $\dfrac{z}{z - e^a}$

(ii) ne^{an} **Ans.** $\dfrac{e^a z}{(z - e^a)^2}$

(iii) $n^2 e^{an}$ **Ans.** $\dfrac{ze^a(z + e^a)}{(z - e^a)^3}$

Exercise 3:

Find the Z-transforms of the following

(i) $e^n \sin 2n$ **Ans.** $\dfrac{ez \sin 2}{z^2 - 2ez\cos 2 + e^2}$

(ii) $c^n \sin n\alpha$ **Ans.** $\dfrac{z(z - c\cos\alpha)}{z^2 - 2cz\cos\alpha + c^2}$

Exercise 4:

Find the Z-transforms of the following

(i) $\cos\left(\frac{n\pi}{2} + \frac{\pi}{4}\right)$ **Ans.** $\dfrac{z(z-1)}{\sqrt{2}(z^2+1)}$

(ii) $\cosh\left(\frac{n\pi}{2} + \theta\right)$ **Ans.** $\dfrac{z^2\cosh\theta - z\cosh\left(\frac{\pi}{2} - \theta\right)}{z^2 - 2z\cosh(\frac{\pi}{2}) + 1}$

Exercise 5:

(a) $\frac{1}{(n+2)!}$ Ans. $z^2(e^{\frac{1}{z}} - 1 - z^{-1})$

(b) $\cosh n\theta$ Ans. $\frac{z(z-\cosh\theta)}{z^2-2z\cosh\theta+1}$

(c) $a^n \sinh\theta$ Ans. $\frac{az\cosh\theta}{z^2-2az\cosh\theta+a^2}$

(d) $\frac{e^{-a}a^n}{n!}$ Ans. $e^{a(z^{-1}-1)}$

(e) $\sin(n+1)\theta$ Ans. $\frac{z^2\sin\theta}{z^2-2z\cos\theta+1}$

(f) $\frac{1}{n+1}$ Ans. $z\ln\left(\frac{z}{z+1}\right)$

Exercise 6:

Find u_2, u_3 if $U(z) = \frac{2z^2+5z+14}{(z-1)^4}$. Ans. $u_0 = 0,\ u_1 = 0, u_2 = 2,\ u_3 = 13$

Exercise 7:

Find u_2, u_3 when $U(z) = \frac{(5z^2+3z+12)}{(z-1)^4}$. Ans. $u_0 = 0,\ u_1 = 0, u_2 = 5,\ u_3 = 23$

Exercise 8:

Determine u_2 where $U(z) = \frac{(2z^2+3z+4)}{(z-3)^3}$ Ans. $u_0 = 0,\ u_1 = 2,\ u_2 = 21$

Example

Find the inverse Z-transform of

(a) $\ln\left(\frac{z}{z+1}\right)$

(b) $\frac{z}{(z+1)^2}$

Solution

(a)

$$\ln\left(\frac{z}{z+1}\right) = -\ln\left(\frac{z+1}{z}\right) = -\ln\left(1+\frac{1}{z}\right)$$

$$= -\left[\frac{1}{z} - \frac{1}{2z^2} + \frac{1}{3z^3} + \cdots\right]$$

$$= \sum_{n=1}^{\infty} \frac{(-1)^n}{n} z^{-n}$$

$$= \sum_{n=0}^{\infty} u_n z^{-n} \qquad \text{where } u_n = \begin{cases} 0 & \text{if } n = 0 \\ \frac{(-1)^n}{n} & \text{if } n > 0 \end{cases}$$

$$\implies \ln\left(\frac{z}{z+1}\right) = Z(u_n)$$

Taking inverse Z-transform of both sides yields

$$Z^{-1}\left[\ln\left(\frac{z}{z+1}\right)\right] = u_n = \begin{cases} 0 & \text{if } n = 0 \\ \frac{(-1)^n}{n} & \text{if } n > 0 \end{cases}.$$

(b)

$$\begin{aligned}
\frac{z}{(z+1)^2} &= \frac{1}{z}\left(1+\frac{1}{z}\right)^{-2} \\
&= \frac{1}{z}\left[1+\frac{(-2)}{z}+\frac{(-2)(-2-1)}{2!z^2}+\frac{(-2)(-2-1)(-2-2)}{3!z^3}+\cdots\right. \\
&= \frac{1}{z}-\frac{2}{z^2}+\frac{3}{z^3}-\frac{4}{z^3}+\cdots \\
&= \sum_{n=0}^{\infty}(-1)^{n-1}nz^{-n} \\
\implies \frac{z}{(z+1)^2} &= Z[(-1)^{n-1}n]
\end{aligned}$$

Taking inverse Z-transform of both sides, we get

$$Z^{-1}\left(\frac{z}{(z+1)^2}\right) = (-1)^{n-1}n.$$

Solution cont. . .

Another Solution: We know that

$$Z[(-1)^n] = \frac{z}{z+1}$$

Therefore

$$Z[(-1)^n n] = -z\frac{d}{dz}\left(\frac{z}{z+1}\right) = -z\left[\frac{z+1-z}{(z+1)^2}\right] = -\frac{z}{(z+1)^2}$$

$$\implies Z[(-1)^n n] = -\frac{z}{(z+1)^2}$$

Taking inverse Z-transform both sides, we have

$$Z^{-1}\left(\frac{z}{(z+1)^2}\right) = (-1)^{n-1} n.$$

Finding inverse Z-transform using partial fraction method

This method is similar to that of finding the inverse Laplace transforms using partial fractions. The method consists of decomposing $\frac{U(z)}{z}$ into partial fractions, multiplying the resulting expansion by z and then inverting the same.

Example

Find the inverse Z-transform of $U(z)$ given by

(a) $U(z) = \dfrac{z}{(Z+4)(z+5)}$

(b) $U(z) = \dfrac{7z - 11z^2}{(z-1)(z-2)(z-3)}$

Example

Find the inverse Z-transform of $U(z)$ given by

(a) $U(z) = \dfrac{z}{(Z+4)(z+5)}$ (b) $U(z) = \dfrac{7z - 11z^2}{(z-1)(z-2)(z-3)}$

Solution

We resolve $\frac{U(z)}{z}$ into partial fraction first.

(a)

$$\frac{U(z)}{z} = \frac{1}{(Z+4)(z+5)} = \frac{1}{z+4} - \frac{1}{z+5}.$$

Hence

$$U(z) = \frac{z}{z+4} - \frac{z}{z+5}$$

Therefore

$$Z^{-1}(U(z)) = Z^{-1}\left(\frac{z}{z+4}\right) - Z^{-1}\left(\frac{z}{z+5}\right) = (-4)^n - (-5)^n$$

Solution cont...

(b)

$$\frac{U(z)}{z} = \frac{7-11z}{(z-1)(z-2)(z-3)} = -\frac{2}{z-1} + \frac{15}{z-2} - \frac{13}{z+3}.$$

Hence

$$U(z) = -\frac{2z}{z-1} + \frac{15z}{z-2} - \frac{13z}{z-3}$$

Therefore

$$\begin{aligned} Z^{-1}(U(z)) &= -2Z^{-1}\left(\frac{z}{z-1}\right) + 15Z^{-1}\left(\frac{z}{z-2}\right) - 13Z^{-1}\left(\frac{z}{z-3}\right) \\ &= -2 + 15(2)^n - 13(3)^n \end{aligned}$$

Example

Find the inverse Z-transform of $\dfrac{4z^2 - 2z}{z^3 - 5z^2 + 8z - 4}$.

Example

Find the inverse Z-transform of $\dfrac{4z^2-2z}{z^3-5z^2+8z-4}$.

Solution

Consider

$$U(z)=\frac{4z^2-2z}{z^3-5z^2+8z-4}=2z\cdot\frac{2z-1}{(z-1)(z-2)^2}\implies\frac{U(z)}{z}=\frac{2(2z-1)}{(z-1)(z-2)^2}.$$

By partial fractions

$$\frac{U(z)}{z}=\frac{2z-1}{z^3-5z^2+8z-4}=\frac{A}{z-1}+\frac{B}{z-2}+\frac{C}{(z-2)^2}$$
$$\implies 2(2z-1)=A(z-2)^2+B(z-1)(z-2)+C(z-1)$$

Putting $z=1$ gives $A=2$ and putting $z=2$ gives $C=6$. Comparing the constant term on both sides of above expression yields

$$-1=4A+2B-C\implies B=-2$$

Solution cont...

Therefore

$$\frac{U(z)}{z} = \frac{2}{z-1} - \frac{2}{z-2} + \frac{6}{(z-2)^2}$$

$$\implies U(z) = \frac{2z}{z-1} - \frac{2z}{z-2} + \frac{6z}{(z-2)^2}$$

Thus

$$Z^{-1}\left[\frac{4z^2-2z}{z^3-5z^2+8z-4}\right] = Z^{-1}\left[\frac{2z}{z-1} - \frac{2z}{z-2} + \frac{6z}{(z-2)^2}\right]$$

$$= 2Z^{-1}\left[\frac{z}{z-1}\right] - 2Z^{-1}\left[\frac{z}{z-2}\right] + 3Z^{-1}\left[\frac{2z}{(z-2)^2}\right]$$

$$= 2 \cdot 1 - 2 \cdot 2^n + 6n2^n$$

$$= 2 - 2^{n+1} + 3n2^n$$

Since $Z(n2^n) = \frac{2z}{(z-2)^2}$.

Example

Find the inverse Z-transform of $\dfrac{z^2}{z^2+1}$.

Example

Find the inverse Z-transform of $\dfrac{z^2}{z^2+1}$.

Solution

Let

$$U(z) = \frac{z^2}{z^2+1}$$

$$\implies \frac{U(z)}{z} = \left[\frac{z}{z^2+1}\right] = \left[\frac{z}{(z+i)(z-i)}\right] = \frac{1}{2}\left[\frac{1}{z+i} + \frac{1}{z-i}\right]$$

$$\implies U(z) = \frac{1}{2}\left[\frac{z}{z+i} + \frac{z}{z-i}\right]$$

Therefore

$$Z^{-1}(U(z)) = Z^{-1}\left(\frac{1}{2}\left[\frac{z}{z+i} + \frac{z}{z-i}\right]\right) = \frac{1}{2}\left[Z^{-1}\left(\frac{z}{z+i}\right) + Z^{-1}\left(\frac{z}{z-i}\right)\right]$$

$$= \frac{(-i)^n + i^n}{2}$$

Example

Find the inverse Z-transform of $\dfrac{2z}{(z-1)(z^2+1)}$.

Example

Find the inverse Z-transform of $\dfrac{2z}{(z-1)(z^2+1)}$.

Solution

Let

$$U(z) = \frac{2z}{(z-1)(z^2+1)} \implies \frac{U(z)}{z} = \frac{2}{(z-1)(z^2+1)}$$

By partial fractions

$$\frac{2}{(z-1)(z^2+1)} = \frac{A}{z-1} + \frac{Bz+C}{z^2+1}$$

$$2 = A(z^2+1) + (Bz+C)(z-1)$$

Putting $z = 1$ gives $A = 1$. Rearranging the similar terms, we have

$$2 = (A+B)z^2 + (-B+C)z + A - C$$

Comparing the coefficients of Z^2 gives $A + B = 0 \implies B = -1$. Comparing the coefficients of z gives $-B + C = 0 \implies C = -1$.

Solution cont...

Therefore

$$\frac{U(z)}{z} = \frac{2}{(z-1)(z^2+1)} = \frac{1}{z-1} - \frac{z+1}{z^2+1}$$

Thus

$$\begin{aligned}
U(z) &= \frac{z}{z-1} - z \cdot \frac{z+1}{z^2+1} \\
&= \frac{z}{z-1} - z \cdot \left[\frac{z}{z^2+1} + \frac{1}{z^2+1} \right] \\
&= \frac{z}{z-1} - z \cdot \left[\frac{z}{(z+i)(z-i)} + \frac{1}{(z+i)(z-i)} \right] \\
&= \frac{z}{z-1} - z \cdot \left[\frac{1}{2} \left(\frac{1}{z+i} + \frac{1}{z-i} \right) + \frac{1}{2i} \left(\frac{1}{z+i} - \frac{1}{z-i} \right) \right] \\
&= \frac{z}{z-1} - \frac{1}{2} \left(\frac{z}{z+i} \right) - \frac{1}{2} \left(\frac{z}{z-i} \right) - \frac{1}{2i} \left(\frac{z}{z+i} \right) + \frac{1}{2i} \left(\frac{z}{z-i} \right)
\end{aligned}$$

Solution cont...

Therefore

$$Z^{-1}(U(z)) = Z^{-1}\left[\frac{z}{z-1} - \frac{1}{2}\left(\frac{z}{z+i}\right) - \frac{1}{2}\left(\frac{z}{z-i}\right) - \frac{1}{2i}\left(\frac{z}{z+i}\right) + \frac{1}{2i}\left(\frac{z}{z-i}\right)\right]$$

$$= Z^{-1}\left(\frac{z}{z-1}\right) - \frac{1}{2}Z^{-1}\left(\frac{z}{z+i}\right) - \frac{1}{2}Z^{-1}\left(\frac{z}{z-i}\right) - \frac{1}{2i}Z^{-1}\left(\frac{z}{z+i}\right) + \frac{1}{2i}Z^{-1}\left(\frac{z}{z-i}\right)$$

$$= 1 - \frac{1}{2}(-i)^n - \frac{1}{2}i^n - \frac{1}{2i}(-i)^n + \frac{1}{2i}i^n$$

$$= 1 - \frac{(-i)^n + i^n}{2} - \frac{(-i)^n - i^n}{2i}$$

Exercise 9:

Find the inverse Z-transform of

(a) $U(z) = \dfrac{z^3 + 2z^2 + 29z}{(z-1)(z+3)^2}$ **Ans.** $Z^{-1}(U(z)) = -n(-3)^n$

(b) $V(z) = \dfrac{2z^2 + 3z}{(z+2)(z+4)}$ **Ans.** $Z^{-1}(V(z)) = \frac{11}{6}(4^n) + \frac{1}{6}(-2)^n$

(c) $F(z) = \dfrac{2z^2 + 3z}{(z+2)(z-4)}$ **Ans.** $Z^{-1}(F(z)) = \dfrac{1}{6}(-2)^n + \dfrac{11}{6}4^n$

(d) $G(z) = \dfrac{z^3 - 20z}{(z-2)^3(z-4)}$ **Ans.** $Z^{-1}(G(z)) = 2^{n-1} + n^2 2^n - 4^n$

Exercise 10:

Find the inverse Z-transform of $\dfrac{1}{(z-5)^3}$.

Ans. $Z^{-1}\left(\dfrac{1}{(z-5)^3}\right) = \begin{cases} \frac{n(n-2)5^{n-3}}{2} & \textbf{if} \quad n \geq 3 \\ 0 & \textbf{if} \quad n < 3 \end{cases}$

Convolution

Convolution

If $\{u_n\}$ and $\{v_n\}$ are two sequences then their convolution is denoted by $u_n * v_n$ and is defined as

$$\{u_n * v_n\} = \sum_{k=0}^{n} u_k v_{n-k} = u_0 v_n + u_1 v_{n-1} + \cdots + u_n v_0$$

- $\{u_n * v_n\} = \{v_n * u_n\}$
- $\{(u_n * v_n) * w_n\} = \{u_n * w_n\} + \{v_n * w_n\}$
- $\{u_n * v_n\} * \{w_n\} = \{u_n\} * \{v_n * w_n\}$

Convolution Theorem

Convolution theorem

If $\{u_n\}$ and $\{v_n\}$ are two sequences such that $Z(u_n)$ and $Z(v_n)$ exists. Then

$$Z(u_n * v_n) = Z(u_n)Z(v_n)$$

In other words, if $Z(u_n) = U(z)$ and $Z(v_n) = V(z)$ then

$$Z^{-1}[U(z)V(z)] = u_n * v_n$$

Example

Use convolution theorem to evaluate $Z^{-1}\left[\frac{z^2}{(z-a)(z-b)}\right]$. Also, find $Z(a^n * a^n)$.

Example

Use convolution theorem to evaluate $Z^{-1}\left[\frac{z^2}{(z-a)(z-b)}\right]$. Also, find $Z(a^n * a^n)$.

Solution

We know that $Z^{-1}\left(\frac{z}{z-a}\right) = a^n$ and $Z^{-1}\left(\frac{z}{z-b}\right) = b^n$. Therefore by convolution theorem

$$Z^{-1}\left[\frac{z^2}{(z-a)(z-b)}\right] = a^n * b^n$$

$$= \sum_{k=0}^{k} a^k b^{n-k}$$

$$= b^n \sum_{k=0}^{k} a^k b^{-k} = b^n \sum_{k=0}^{n} (ab^{-1})^n$$

$$= b^n \left[\frac{1-(ab^{-1})^{n+1}}{1-ab^{-1}}\right] = \frac{a^{n+1}-b^{n+1}}{a-b}$$

Solution cont. . .

Further,

$$a^n * a^n = \sum_{k=0}^{\infty} a^k a^{n-k} = a^n \sum_{k=0}^{n} 1 = (n+1)a^n$$

Here $Z(a^n) = \frac{z}{z-a}$. Therefore

$$Z(a^n * a^n) = Z(a^n)(a^n) = \frac{z^2}{(z-a)^2}$$

If we put the value of $a^n * a^n$ then we get

$$Z[(n+1)a^n] = \frac{z^2}{(z-a)^2}$$

Example

(a) Applying convolution, show that $\left\{\frac{1}{n!} * \frac{1}{n!}\right\} = \left\{\frac{2^n}{n!}\right\}$.

(b) Verify convolution theorem for $\{u_n\} = \{n\}$ and $\{v_n\} = \{n^2\}$.

Solution

(a) By definition of convolution

$$\left\{\frac{1}{n!} * \frac{1}{n!}\right\} = \sum_{k=0}^{n} \frac{1}{k!}\frac{1}{(n-k)!}$$

$$= \frac{1}{n!} + \frac{1}{1!(n-1)!} + \frac{1}{2!(n-2)!} + \cdots + \frac{1}{n!}$$

$$= \frac{1}{n!} + \frac{n}{1!n!} + \frac{n(n-1)}{2!n!} + \cdots + \frac{n!}{n!n!}$$

$$= \frac{1}{n!}\left[1 + \frac{n}{1!} + \frac{n(n-1)}{2!} + \cdots + 1\right]$$

$$= \frac{1}{n!}\left[{}^{n}C_0 + {}^{n}C_1 + {}^{n}C_2 + \cdots + {}^{n}C_n\right]$$

$$= \frac{(1+1)^n}{n!}$$

$$= \frac{2^n}{n!}$$

Solution cont...

(b) Here we have $\{u_n\} = \{n\}$ and $\{v_n\} = \{n^2\}$, therefore

$$u_n * v_n = \sum_{k=0}^{n} k^2(n-k) = n\sum_{n=0}^{n} k^2 - \sum_{k=0}^{n} k^3$$
$$= n\left[\frac{n(n+1)(2n+1)}{6}\right] - \frac{n^2(n+1)^2}{4} = \frac{n^4 - n^2}{12}$$

Thus

$$Z(n^2 * n) = \frac{1}{12}Z(n^4) - \frac{1}{12}Z(n^2)$$
$$= \frac{1}{12}\frac{z(z^3 + 11z^2 + 11z + 1)}{(z-1)^5} - \frac{1}{12} - \frac{1}{12}\frac{z^2 + z}{(z-1)^3} = \frac{z^3 + z^2}{(z-1)^5}$$

Also, we have

$$Z(n^2)Z(n) = \frac{z^2 + z}{(z-1)^3} \cdot \frac{z}{(z-1)^2} = \frac{z^3 + z^2}{(z-1)^5}$$

Thus, convolution theorem $Z(n^2 * n) = Z(n^2)Z(n)$ has been verified.

Exercise 11:

Using convolution theorem, find the inverse Z-transform of $\left(\frac{z}{z-a}\right)^3$. Deduce for $\left(\frac{z}{z-1}\right)^3$.

Hint: Apply the convolution theorem twice.

Ans. $Z^{-1}\left[\left(\frac{z}{z-a}\right)^3\right] = \frac{n(n+1)}{2}a^n$ **and** $Z^{-1}\left[\left(\frac{z}{z-1}\right)^3\right] = \frac{n(n+1)}{2}$

Exercise 12:

Using the convolution theorem, find the Z-transform of the following

(a) $\frac{z^2}{(z-1)(z-3)}$

(b) $\left(\frac{z}{z-a}\right)^2$.